# Alexander (Sasha) Apartsin, Ph.D.


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## SELECTED RECOGNITION & ACHIEVEMENTS

**AI Product of the Year** recognition for SafeRide automotive AI product work.

**$250K Citi LabLink grant** for a financial data science collaboration platform.

**IIA innovation grants** for Motorola Solutions industry-academia AI projects.

**$10M hardware-cost savings contribution** through model-based capacity planning at Amdocs.

## INNOVATION & ENTREPRENEURSHIP

**Founder / CTO experience** across search, multimedia, cloud optimization, and applied AI ventures; corporate innovation through AI labs, grants, patents, and prototypes.

## RESEARCH & IMPACT

**Publications:** 40+ Scholar-indexed works and 470+ citations across ML, vision, NLP, deep learning, and signal processing.

**Patents:** 10 granted/filed across vehicle diagnostics, driver monitoring, model configuration, onboard inference, public-safety audio, and search/image matching.

## TEACHING & MENTORSHIP

**24+ supervised B.Sc. / M.Sc. final projects** since 2012, with teaching across HIT, Afeka, TAU, and BIU; multi-course, project-based series in Language, Vision, and Scalable AI, each with a companion online textbook, delivered over 16 past offerings with 112+ student course projects.

## AI/ML DOMAINS

Vision | Audio | Speech | Text / NLP | LLMs | Diffusion | Task Allocation | Anomaly Detection | RecSys | Time Series / Signals | Novel Material Discovery | Big Data

## INDUSTRIES

Automotive | Finance | Public Safety | Healthcare | Telecom | Media / Multimedia | Document AI | Education | Search | Industrial Inspection

## HANDS-ON

Python | PyTorch | Hugging Face | SparkNLP / spaCy | OpenCV | LangChain / LangGraph | OpenAI | C / C++ | SQL | Pandas / NumPy | Databricks | MLflow | Horovod | AWS / Azure

- **AI researcher and technology leader** with 30+ years of experience across machine learning, signal processing, computer vision, NLP, and generative AI.
- **End to end, from research to product:** builds and leads research and engineering teams, drives cross-functional collaboration, co-defines products with customers, and executes delivery from original research and algorithms through production systems to shipped products, staying hands-on alongside the team.
- **Academic and applied impact** through research, teaching, and student supervision, with applications across healthcare, automotive, public safety, finance, and enterprise AI.

## LEADERSHIP & RESEARCH EXPERIENCE

**Faculty Member** - HIT, School of Computer Science — **2024-Present**

Conduct applied AI research in generative AI, large language models, vision, and signals; created hands-on, project-based course series on LLMs, GenAI, and diffusion models taught across HIT (School of Computer Science and Digital Healthcare Department) and Bar-Ilan University (undergraduate and graduate programs); supervise M.Sc. students in Afeka College of Engineering's Intelligent Systems program.

**Head of Data Science** - Questar / SafeRide Technologies — **2020-2024**

Built and hired the data science team; led big-data AI for vehicle telematics across signal, audio, graph, anomaly detection, and onboard-model work; led OEM-partner projects including engine-health monitoring (Ford), steering-by-wire monitoring (Hyundai Mobis), in-cabin acoustic fault detection (Renault), fuel-consumption modeling (Aptiv), and fault-sequence analysis (M.A.N.); engaged directly with customers to shape requirements; led the definition of a no-code product for onboard monitoring models.

**VP of Artificial Intelligence** - harmon.ie — **2019-2020**

Led AI for an enterprise document-management product designed to help organizations handle the growing amount of information in Office 365; managed a cross-functional data-science and engineering team and shaped the research agenda across large-scale text understanding, user-behavior modeling from Office-graph signals, and recommendation under limited supervision.

**Head of AI Research** - Motorola Solutions, Israel Innovation Center — **2015-2019**

Built and led the research team at Motorola Solutions' first AI lab in the Innovation Center; designed and led execution of the roadmap across video, text, speech, and public-safety AI; established internal cross-functional partnerships and academia-industry collaborations funded by Israel Innovation Authority programs and binational grants.

**AVP Data Science** - Citi Israel Innovation Lab — **2013-2015**

Led financial data-science projects: bond-price prediction for auctions, FX causality analysis and trader-informativeness classification, anomaly detection for trader compliance auditing, and trade-to-wire matching via correlation and constraint optimization.

**Founder / CTO / R&D and algorithm leadership** - SearchNG, CloudTuner, MPCode, Tegrity, AppiVision, Amdocs, Comverse, earlier roles — **1991-2012**

Founded ventures and led hands-on intelligent-system work in search, multimedia, video streaming, wireless communications, telecom, predictive modeling, handwriting recognition, imaging, inspection, and signal acquisition.

## EDUCATION

**Postdoctoral Fellowship** - Weizmann Institute of Science · computer vision and machine learning — **2013**

**Ph.D., Computer Science** - Tel Aviv University · machine learning and signal processing — **2012**

**M.Sc., Technology Management** - Polytechnic Institute of NYU — **2006**

**M.Sc., Computer Science** - Weizmann Institute of Science · computer vision — **2004**

**B.Sc., Computer Science** - Technion — **1995**